

PlantVillage 38-Class Crop Disease Detection

Model Training & Evaluation Report

54,304

Total images

38

Disease / healthy classes

99.77%

Test Macro-F1

This report consolidates the supplied PlantVillage dataset statistics, class distribution, class-weight configuration, 15-epoch training log, learning-rate changes, validation results, test-set classification report, and the supplied training curves and 38-class confusion matrix.

Headline results

Best validation Macro-F1: **99.93%** at epoch 15. Final reported test accuracy: **99.80%**; test Macro-F1: **99.77%**; weighted F1: **99.80%**.

Source: user-supplied dataset metadata, training log, classification report, and figures.

1. Dataset Overview

The supplied dataset contains 54,304 images across 38 PlantVillage classes. Each sample includes image data plus metadata fields including class_label, class_idx, host, disease, split, leaf_id, and leaf_grouped. The supplied example is an RGB 256×256 image.

Split	Images	Share
Train	36,852	67.86%
Validation	6,504	11.98%
Test	10,948	20.16%
Total	54,304	100.00%

Class Distribution and Class Weights

Idx	Class	Train	Val	Test	Weight
0	Apple / Apple scab	431	76	123	2.250
1	Apple / Black rot	422	75	124	2.298
2	Apple / Cedar apple rust	187	33	55	5.186
3	Apple / healthy	1123	198	324	0.864
4	Blueberry / healthy	1040	184	278	0.932
5	Cherry (including sour) / Powdery mildew	680	120	252	1.426
6	Cherry (including sour) / healthy	574	101	179	1.690
7	Corn (maize) / Cercospora leaf spot Gray leaf spot	349	61	103	2.779
8	Corn (maize) / Common rust	811	143	238	1.196
9	Corn (maize) / Northern Leaf Blight	670	118	197	1.447
10	Corn (maize) / healthy	790	140	232	1.228
11	Grape / Black rot	802	142	236	1.209
12	Grape / Esca (Black Measles)	942	166	275	1.030
13	Grape / Leaf blight (Isariopsis Leaf Spot)	731	129	216	1.327
14	Grape / healthy	287	51	85	3.379
15	Orange / Haunglongbing (Citrus greening)	3698	653	1156	0.262
16	Peach / Bacterial spot	1563	276	458	0.620
17	Peach / healthy	248	44	68	3.910
18	Pepper, bell / Bacterial spot	687	121	189	1.412
19	Pepper, bell / healthy	997	176	304	0.973
20	Potato / Early blight	680	120	200	1.426
21	Potato / Late blight	680	120	200	1.426
22	Potato / healthy	102	18	32	9.508
23	Raspberry / healthy	256	45	70	3.788
24	Soybean / healthy	3480	614	996	0.279
25	Squash / Powdery mildew	1248	220	367	0.777
26	Strawberry / Leaf scorch	727	128	254	1.334
27	Strawberry / healthy	309	55	92	3.138
28	Tomato / Bacterial spot	1448	255	424	0.670
29	Tomato / Early blight	680	120	200	1.426
30	Tomato / Late blight	1298	229	382	0.747
31	Tomato / Leaf Mold	646	114	192	1.501
32	Tomato / Septoria leaf spot	1199	212	360	0.809
33	Tomato / Spider mites Two-spotted spider mite	1139	201	336	0.851
34	Tomato / Target Spot	955	168	281	1.015
35	Tomato / Tomato Yellow Leaf Curl Virus	3645	643	1069	0.266
36	Tomato / Tomato mosaic virus	253	45	75	3.833
37	Tomato / healthy	1075	190	326	0.902

2. Training Progress

Training was run for 15 epochs with 576 training batches and 102 validation batches per epoch. The supplied log records loss, accuracy, Macro-F1, and learning rate. The best checkpoint was updated whenever validation Macro-F1 improved.

Epoch-by-Epoch Results

Ep	Train Loss	Train Acc	Train F1	Val Loss	Val Acc	Val F1	LR
1	0.9280	92.66%	90.53%	0.6558	98.46%	97.46%	3.00e-04
2	0.6512	98.73%	98.31%	0.6163	99.55%	99.27%	3.00e-04
3	0.6337	99.09%	98.68%	0.6103	99.55%	99.40%	3.00e-04
4	0.6213	99.38%	99.02%	0.6164	99.11%	98.73%	3.00e-04
5	0.6173	99.40%	99.22%	0.6073	99.58%	99.44%	3.00e-04
6	0.6169	99.40%	99.21%	0.6030	99.57%	99.30%	3.00e-04
7	0.6082	99.59%	99.49%	0.6054	99.57%	99.43%	3.00e-04
8	0.6147	99.38%	99.03%	0.6189	99.25%	98.84%	9.00e-05
9	0.6005	99.75%	99.59%	0.5954	99.71%	99.50%	9.00e-05
10	0.5940	99.90%	99.87%	0.5917	99.83%	99.70%	9.00e-05
11	0.5934	99.91%	99.88%	0.5902	99.91%	99.86%	9.00e-05
12	0.5927	99.94%	99.91%	0.5897	99.94%	99.89%	9.00e-05
13	0.5916	99.92%	99.90%	0.5941	99.86%	99.73%	9.00e-05
14	0.5942	99.90%	99.83%	0.5927	99.86%	99.77%	9.00e-05
15	0.5919	99.93%	99.90%	0.5870	99.95%	99.93%	9.00e-05

Learning-rate schedule observed in the log: 3.00×10^{-4} through epoch 7, then 9.00×10^{-5} from epoch 8 onward.

Best validation result: epoch 15 with validation loss 0.5870, validation accuracy 99.95%, and validation Macro-F1 99.93%.

3. Training Curves

The supplied figure shows the training/validation loss and training/validation accuracy over the 15 epochs. The curves converge to very high accuracy while loss decreases substantially.

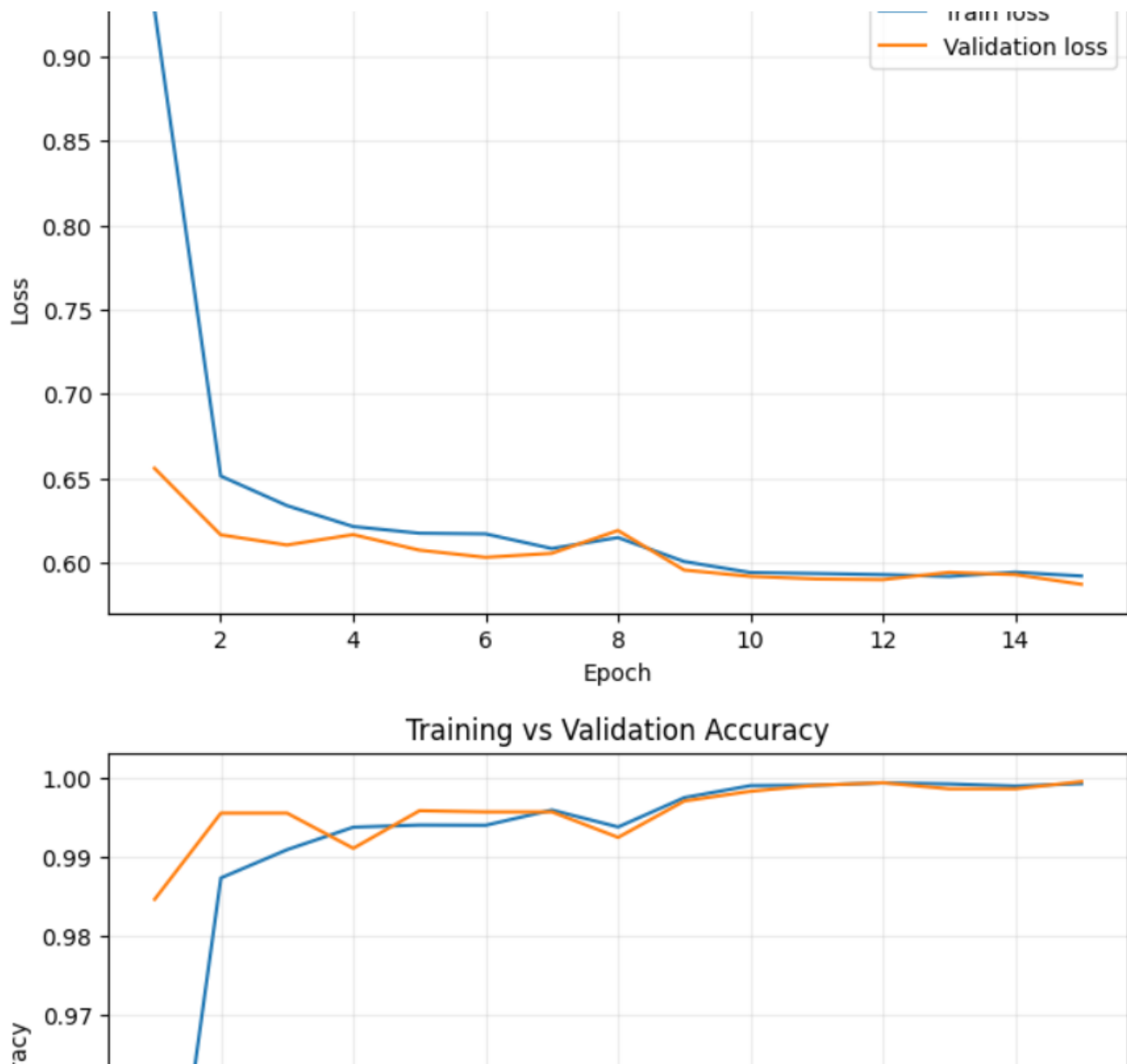


Figure 1. Supplied training vs. validation loss and accuracy curves.

4. Test-Set Evaluation

The supplied classification report evaluates all 10,948 test images across all 38 classes. Precision, recall, F1-score, and support are reproduced below.

Class	Precision	Recall	F1-score	Support
Apple / Apple scab	1.0000	0.9919	0.9959	123
Apple / Black rot	1.0000	1.0000	1.0000	124
Apple / Cedar apple rust	1.0000	1.0000	1.0000	55
Apple / healthy	0.9969	1.0000	0.9985	324
Blueberry / healthy	1.0000	1.0000	1.0000	278
Cherry (including sour) / Powdery mildew	1.0000	1.0000	1.0000	252
Cherry (including sour) / healthy	1.0000	1.0000	1.0000	179
Corn (maize) / Cercospora leaf spot Gray leaf spot	1.0000	0.9612	0.9802	103
Corn (maize) / Common rust	1.0000	1.0000	1.0000	238
Corn (maize) / Northern Leaf Blight	0.9801	1.0000	0.9899	197
Corn (maize) / healthy	1.0000	1.0000	1.0000	232
Grape / Black rot	0.9915	0.9915	0.9915	236
Grape / Esca (Black Measles)	0.9927	0.9927	0.9927	275
Grape / Leaf blight (Isariopsis Leaf Spot)	1.0000	1.0000	1.0000	216
Grape / healthy	1.0000	1.0000	1.0000	85
Orange / Haunglongbing (Citrus greening)	1.0000	1.0000	1.0000	1156
Peach / Bacterial spot	1.0000	1.0000	1.0000	458
Peach / healthy	1.0000	1.0000	1.0000	68
Pepper, bell / Bacterial spot	1.0000	1.0000	1.0000	189
Pepper, bell / healthy	1.0000	1.0000	1.0000	304
Potato / Early blight	1.0000	1.0000	1.0000	200
Potato / Late blight	1.0000	0.9850	0.9924	200
Potato / healthy	1.0000	1.0000	1.0000	32
Raspberry / healthy	1.0000	1.0000	1.0000	70
Soybean / healthy	1.0000	1.0000	1.0000	996
Squash / Powdery mildew	1.0000	1.0000	1.0000	367
Strawberry / Leaf scorch	1.0000	1.0000	1.0000	254
Strawberry / healthy	1.0000	1.0000	1.0000	92
Tomato / Bacterial spot	1.0000	0.9882	0.9941	424
Tomato / Early blight	0.9900	0.9950	0.9925	200
Tomato / Late blight	0.9921	0.9895	0.9908	382
Tomato / Leaf Mold	1.0000	1.0000	1.0000	192
Tomato / Septoria leaf spot	0.9972	1.0000	0.9986	360
Tomato / Spider mites Two-spotted spider mite	0.9941	1.0000	0.9970	336
Tomato / Target Spot	1.0000	1.0000	1.0000	281
Tomato / Tomato Yellow Leaf Curl Virus	0.9963	1.0000	0.9981	1069
Tomato / Tomato mosaic virus	1.0000	1.0000	1.0000	75
Tomato / healthy	0.9969	1.0000	0.9985	326

Overall test metrics: accuracy = 99.80%, macro-average precision = 99.81%, macro-average recall = 99.72%, macro-average F1 = 99.77%, and weighted-average precision/recall/F1 are each approximately 99.80% as supplied.

5. 38-Class Confusion Matrix

The supplied confusion matrix compares true classes (rows) against predicted classes (columns). The strong concentration along the diagonal visually indicates very few test-set misclassifications.

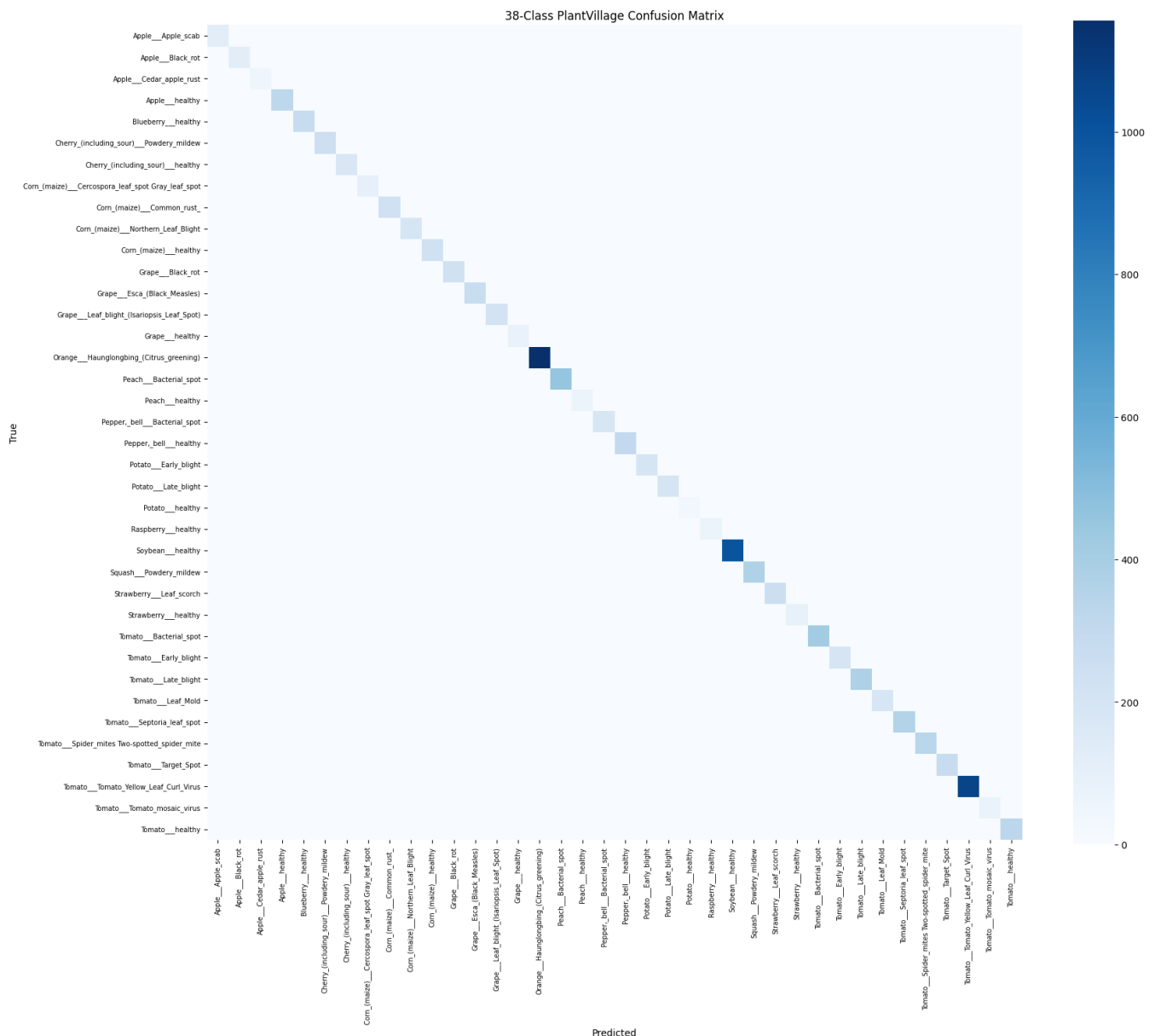


Figure 2. Supplied 38-class PlantVillage confusion matrix.

6. Key Findings

- The model improved rapidly during the first few epochs: validation accuracy rose from 98.46% in epoch 1 to 99.55% in epoch 2. Validation Macro-F1 reached 99.40% by epoch 3.
- The strongest validation result occurred at epoch 15: 99.95% accuracy and 99.93% Macro-F1, with validation loss 0.5870.
- The final supplied test report gives 99.80% accuracy and 99.77% Macro-F1 over 10,948 test images.
- The lowest class-level test F1-scores in the supplied report are still high: Corn Cercospora/Gray Leaf Spot 98.02%, Potato Late Blight 99.24%, Grape Black Rot 99.15%, and Tomato Late Blight 99.08%.
- Class weights were used to address the substantial class imbalance. The supplied training counts range from 102 images for Potato healthy to 3,698 images for Orange Huanglongbing.

Important scope note: These metrics describe performance on the supplied PlantVillage split. They do not by themselves establish performance on field photographs or other out-of-distribution images.

Final Model Selection

Selection criterion	Result
Best epoch	15
Best validation Macro-F1	99.93%
Validation accuracy at best epoch	99.95%
Validation loss at best epoch	0.5870
Reported test accuracy	99.80%
Reported test Macro-F1	99.77%

Appendix A — Supplied Training Log

The complete epoch-level training log is included below in compact form, preserving the supplied values. This appendix is intended for reproducibility and audit/reference.

Epoch	Training result	Validation result	LR	Checkpoint status
1	Loss 0.9280 Acc 92.66% F1 90.53%	Loss 0.6558 Acc 98.46% F1 97.46%	3.00e-04	NEW BEST MODEL
2	Loss 0.6512 Acc 98.73% F1 98.31%	Loss 0.6163 Acc 99.55% F1 99.27%	3.00e-04	NEW BEST MODEL
3	Loss 0.6337 Acc 99.09% F1 98.68%	Loss 0.6103 Acc 99.55% F1 99.40%	3.00e-04	NEW BEST MODEL
4	Loss 0.6213 Acc 99.38% F1 99.02%	Loss 0.6164 Acc 99.11% F1 98.73%	3.00e-04	No improvement
5	Loss 0.6173 Acc 99.40% F1 99.22%	Loss 0.6073 Acc 99.58% F1 99.44%	3.00e-04	NEW BEST MODEL
6	Loss 0.6169 Acc 99.40% F1 99.21%	Loss 0.6030 Acc 99.57% F1 99.30%	3.00e-04	No improvement
7	Loss 0.6082 Acc 99.59% F1 99.49%	Loss 0.6054 Acc 99.57% F1 99.43%	3.00e-04	No improvement
8	Loss 0.6147 Acc 99.38% F1 99.03%	Loss 0.6189 Acc 99.25% F1 98.84%	9.00e-05	No improvement
9	Loss 0.6005 Acc 99.75% F1 99.59%	Loss 0.5954 Acc 99.71% F1 99.50%	9.00e-05	NEW BEST MODEL
10	Loss 0.5940 Acc 99.90% F1 99.87%	Loss 0.5917 Acc 99.83% F1 99.70%	9.00e-05	NEW BEST MODEL
11	Loss 0.5934 Acc 99.91% F1 99.88%	Loss 0.5902 Acc 99.91% F1 99.86%	9.00e-05	NEW BEST MODEL
12	Loss 0.5927 Acc 99.94% F1 99.91%	Loss 0.5897 Acc 99.94% F1 99.89%	9.00e-05	NEW BEST MODEL
13	Loss 0.5916 Acc 99.92% F1 99.90%	Loss 0.5941 Acc 99.86% F1 99.73%	9.00e-05	No improvement
14	Loss 0.5942 Acc 99.90% F1 99.83%	Loss 0.5927 Acc 99.86% F1 99.77%	9.00e-05	No improvement
15	Loss 0.5919 Acc 99.93% F1 99.90%	Loss 0.5870 Acc 99.95% F1 99.93%	9.00e-05	NEW BEST MODEL

Training complete. Best validation Macro-F1: 0.9993. Best epoch: 15.

End of report.